

CNN (Convolution Neural Networks)

it is a type of deep neural networks which process the data in grid like topology

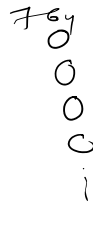
- 1) image classification (e.g. cat, dog)
- 2) object detection (number plates, human in crowd)
- 3) Face recognition
- 4) medical imaging (x-rays)
- 5) self driving car vision.



1) Input layer: Take image input ($\frac{28 \times 28}{764}$)



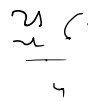
2) Convolution layer: Applies kernel to extract the features like edges, colors, textures



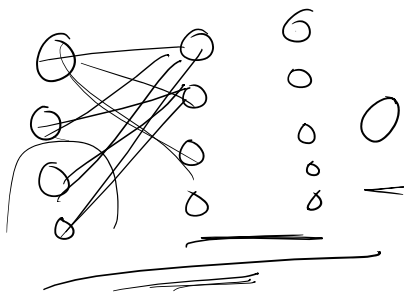
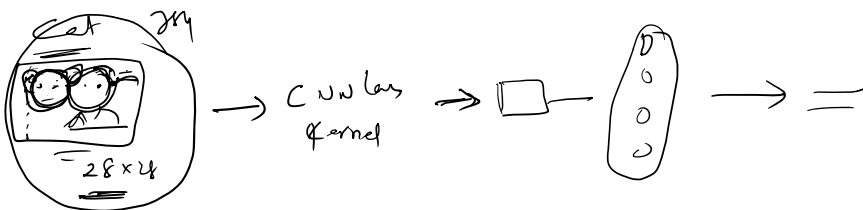
3) Activation function: ReLU, $\max(0, x)$

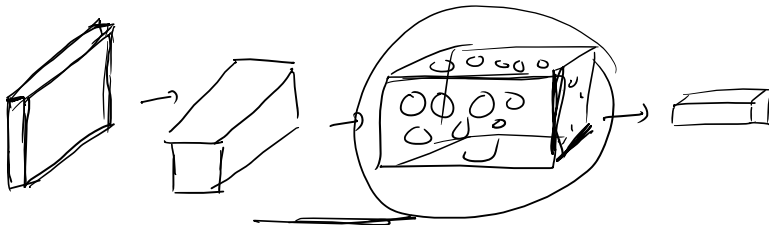
4) Pooling layer: Reduces spatial size

5) Flatten layer convert 2D to 1D

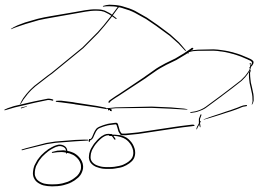
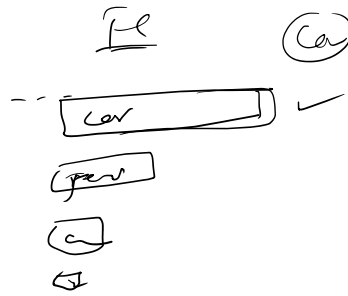


6) Fully Connected layer Find classification

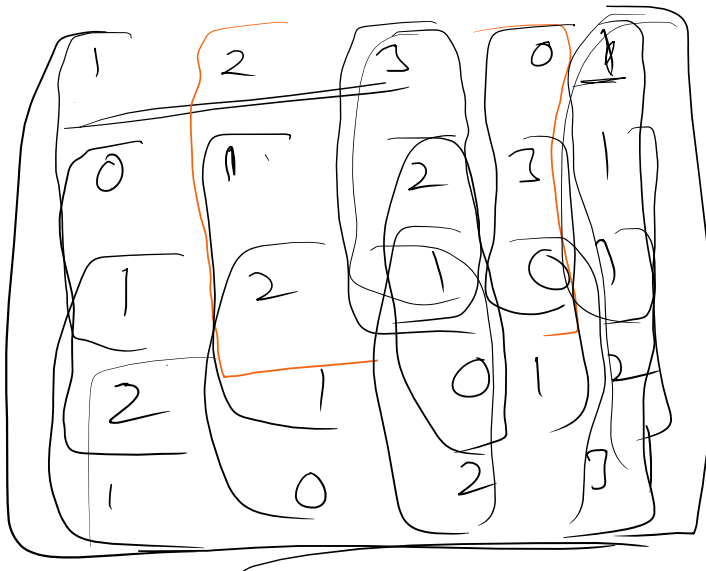




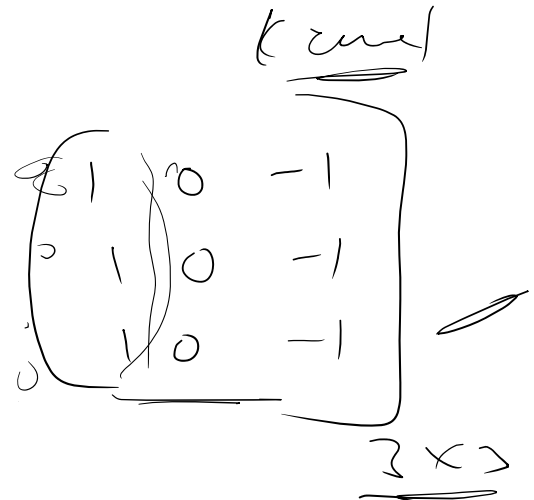
Input $\rightarrow$ Conv $\rightarrow$ ReLU $\rightarrow$ Pool $\rightarrow$ FC



5x5



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00
00



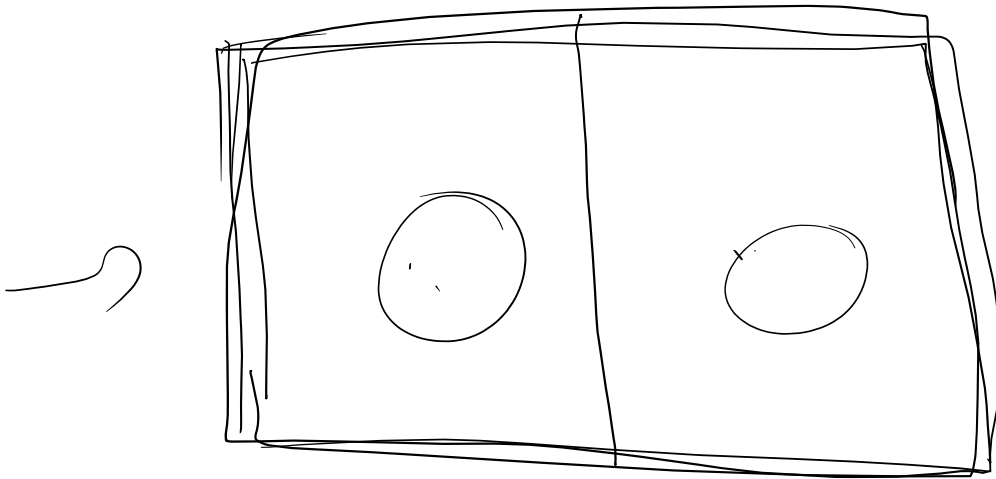
Identical edges
Stride = padding

$$= 1 + 0 + 3 + 0 + 0 - 2 + 1 + 0 - 1 = -4 \quad 3 \quad 0$$

$$= 2 + 0 - 0 + 1 + 0 - 3 + 2 + 1 + 0 =$$

1 2 3 4 5

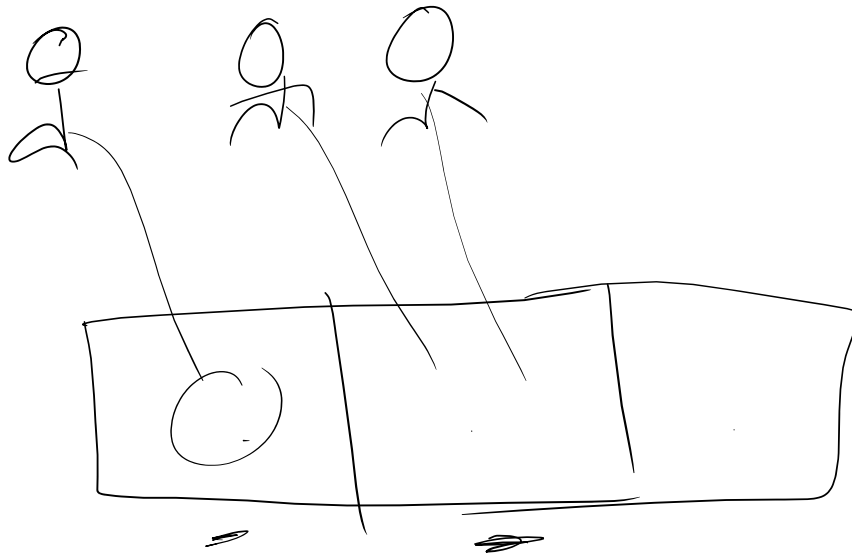
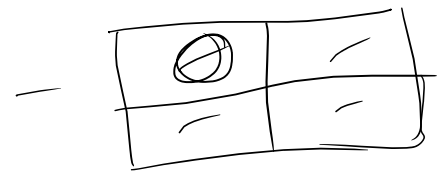
1 2



corn

(2, 2, 1)

corn



1, 3, 1
0, 3, 2
1, 3, 3

or
or
or

① 3, 1	① 3, 2	① 3, 3
② 3, 1	② 3, 2	② 3, 3

2, 3, 1

2, 3, 1	4, 5, 2	1, 3, 5
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